TRIBHUVAN UNIVERSITY

INSTITUTE OF ENGINEERING

THAPATHALI CAMPUS

A Project Proposal

On

चित्रण: An Automated Festive Poster Generator with Wishes in Nepali Language

Submitted By:

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Submitted To:

Department of Electronics and Computer Engineering

Thapathali Campus

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In partial fulfillment for the award of the Bachelor's Degree in Computer Engineering

Under the Supervision of

Er. Saroj Shakya

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ABSTRACT

The regular design of festival posters can be tedious and time-consuming, particularly for Nepalese people where more than 50 festivals are celebrated. This system aims to automate the creation of Nepalese festival posters in Nepali text by integrating Natural Language Processing with Computer Vision. The proposed system accepts a user prompt in English language as input to generate a festival posters containing images and Nepali text. The system works in two phases: first, it converts an English prompt into a Nepali festival wish using a fine-tuned M2M-100 model, and extracts the festival theme. In the second phase, a pre-trained Latent Diffusion Model generates an image based on the festival theme. The generated image's color style is then applied to the text and the text's position is determined through saliency mapping. Finally, the styled text and image are integrated to produce a complete festival poster. This method simplifies the manual process of creating festival posters, enhancing efficiency and creativity. It is a valuable tool for graphic designers and anyone looking to quickly and effectively generate festive posters without prior design knowledge.

Keywords: Computer Vision, Festival poster, Latent Diffusion Model, M2M-100 Model, Natural Language Processing, Nepali Text, Saliency Mapping

Table of Contents

ACKNOWLEDGEMENT ..... **i**
ABSTRACT ..... **ii**
List of Figures ..... **v**
List of Tables ..... **vi**
List of Abbreviations ..... **vii**
1 INTRODUCTION **1**
1.1 Background 1
1.2 Motivation 1
1.3 Problem Definition 2
1.4 Objectives 2
1.5 Scope and Applications 3
1.5.1 Scope 3
1.5.2 Applications 3
2 LITERATURE REVIEW **5**
3 METHODOLOGY **13**
3.1 System Architecture 13
3.1.1 Working Principle 14
3.2 Dataset Preparation 14
3.2.1 Dataset for Title Generation in Nepali Font 14
3.2.2 Dataset for Poster-Generation in Nepali Font 15
3.3 Title Generation 15
3.3.1 M2M-100 Architecture 16
3.3.2 Evaluation Metrics 18
3.4 Image Generation 19
3.4.1 Latent Diffusion Model 20
3.4.2 Evaluation Metrics 24
3.5 Text Styling 25
3.6 Text-Image Integration 26
3.7 Flowchart 27
3.8 Instrumentation Tools 28

3.8.1 System Requirements....28
3.8.2 Software Tools and Library Requirements....28
4 EXPECTED OUTCOME....30
4.1 Visual Quality....30
4.2 Relevance and Accuracy....30
5 PROJECT SCHEDULE....33
6 PROJECT BUDGET....34
7 FEASIBILITY ANALYSIS....35
7.1 Technical Feasibility....35
7.2 Operational Feasibility....35
7.3 Economic Feasibility....35
7.4 Schedule Feasibility....35
7.5 Legal Feasibility....35
References....36

List of Figures

Figure 3-1 System Architecture 13
Figure 3-2 Dataset Preparation for Tihar Festival 15
Figure 3-3 English to Nepali Text with Dense and Language Specific Sparse Model 18
Figure 3-4 Architecture of Latent Diffusion Model 21
Figure 3-5 Text Styling Procedure for Generated Text 26
Figure 3-6 Integration of Generated Image and Styled Text 26
Figure 3-7 System Flowchart 27
Figure 4-1 Poster for Dashain Festival 31
Figure 4-2 Images generated for Respective prompts. 32

List of Tables

Table 2-1 Summary Table of Literature Review 10
Table 5-1: Project Timeline Gantt chart 33
Table 6-1 Budget Estimation 34

List of Abbreviations

ADA	Adaptive Discriminator Augmentation
AdaIN	Adaptive Instance Normalization
AttnGAN	Attention Generative Adversarial Network
AuPoD	Automatic Poster Design
BERT	Bidirectional Encoder Representation from Transformers
BLEU	Bilingual Evaluation Understudy
BP	Brevity Penalty
BPE	Byte Pair Encoding
CGL-GAN	Composition-aware Graphic Layout Generative Adversarial Network
CIFAR	Canadian Institute for Advanced Research
CLIP	Contrastive Language-Image Pre-training
CNN	Convolutional Neural Network
COCO	Common Objects in Context
CSS	Cascading Style Sheet
CSV	Comma-Separated Values
CTR	Click-Through Rate
CUB	Caltech-UCSD Birds
DAMSM	Deep Attention Multimodal Similarity Model
DC-GAN	Deep Convolutional Generative Adversarial Network
DDPM	Denoising Diffusion Probabilistic Model
DF-GAN	Deep Fusion Generative Adversarial Networks
FFHQ	Flickr-Faces-HQ
FFN	Feed Forward Network
FID	Frechet Inception Distance
GANs	Generative Adversarial Networks
GLIDE	Generative Latent Image Disentanglement
GPUs	Graphics Processing Units
GUI	Graphical User Interface
HTML	HyperText Markup Language
ICVT	Image Captioning Visual Transformer
IDE	Integrated Development Environment
IS	Inception Score

KL-Reg	Kullback-Leibler Regularization
LAION	Large-scale Artificial Intelligence Open Network
LDM	Latent Diffusion Model
LLMS	Large Language Model Super-resolution
LSTM	Long Short Term Memory
LSUN	Large-scale Scene UNderstanding challenge
M2M	Many to Many
MA-GP	Mutual Alignment Gradient Penalty
mBART	Multilingual Bidirectional Auto-Regressive Transformers
MFEN	Multimodality Feature Extraction Net
MLP	Multi-Layer Perceptron
MMCQ	Modified Median Cut Quantization
MMT	Multilingual Machine Translation
mT5	Multilingual Text-to-Text Transfer Transformer
NLP	Natural Language Processing
NLTK	Natural Language ToolKit
OCR	Optical Character Recognition
PIL	Python Imaging Library
PNG	Portable Network Graphics
PSD	Photoshop Document
ReLU	Rectified Linear Unit
RGB	Red Green Blue
ROUGE	Recall-Oriented Understudy for Gisting Evaluation
RPM	Revenue per Mile
SAP	Style Attribute Predictor
SOTA	State-of-the-art
t-SNE	t-distributed Stochastic Neighbor Embedding
UCSD	University of California San Diego
VAE	Variational Autoencoders
VGG-16	Visual Geometry Group 16
ViT-S	Vision Transformer Small
VQGAN	Vector Quantized Generative Adversarial Network
VQ-Reg	Vector Quantization Regularization

1 INTRODUCTION

The regular design of festival posters is tedious and time-consuming, particularly for Nepalese people where more than 50 festivals are celebrated. Each festival has its importance and represents the diversity of Nepalese culture. Wishing and celebrating one another festival is a beautiful form of showing respect for each other's culture and tradition. With the increasing popularity of digital platforms, creating festival posters and posting them to celebrate with everyone is trending. However, regularly creating visually appealing festival posters needs some knowledge of design and time. Adding Nepali text makes the process more difficult and time-consuming.

To address this, we are working on a project titled 'चित्रण : An Automated Festive Poster Generator with Wishes in Nepali Language'. Chitran creates beautiful posters with Nepali texts for Nepali festivals (Dashain, Tihar, Chhath, New Year, and Holi). It takes prompts in English and creates visually appealing posters that include festive wishes in Nepali text, based on the prompts given. With Chitran, celebrating and sharing the joy of Nepali festivals becomes simpler and more visually engaging, while keeping the cultural essence in every poster.

1.1 Background

With the advancement of Natural Language Processing, understanding human intention through text prompts and generating a title is possible. Also, generative models help to create quality images and styles related to the festival. Combining NLP with generative models will provide an effective approach to making the festival posters with less user interaction. Previously, research has been done to create posters captions or images using generative models. Some have created posters in other languages aside from English. However, no research has been done to create Nepali festival posters with Nepali text.

1.2 Motivation

In the era of social media, where social media presence and visual communication immensely impact audience engagement, firms, clubs, and organizations are constantly seeking ways to connect with their audiences. In recent years, one of the most common

methods of doing so across multiple domains of Nepali social media accounts has been through festival posters.

However, traditional approaches to poster design present several challenges. Hiring graphics designers for repetitive tasks as such can be costly, tedious, and inefficient along with neglecting resources and time due to its extensive manpower and time investment. Consequently, valuable manpower and funds are diverted to a less prioritized and redundant task which is most likely to harm the performance of the company.

These reasons underscore the need for automation of festival posters, enabling organizations to generate as many festival posters with just a click of a button.

Thus, automation of festival social media posters creation can bring about significant improvement in an organization's performance by redirecting the economy and manpower towards more strategic and impactful activities.

1.3 Problem Definition

The festival social media posters generation is a repetitive task. Using the traditional approach for such redundant tasks results not only in improper allocation of resources, manpower, and economy within the organization ultimately resulting in inefficient performance but also underestimates the potential of the hired professional. Hence, the redundant task of designing each festival's social media posters is alleviated by automating the system. The automated process can generate images and text and integrate them to create a cohesive and aesthetically pleasing posters.

1.4 Objectives

- To analyze input prompt to extract event and year details, then generate concise Nepali short title.
- To create Nepali festival-themed image and integrate it with styled titles in Nepali to create a digital poster.

1.5 Scope and Applications

With the help of generative models and natural language processing, this project attempts to create an automated system that, for small enterprises, graphic artists, and digital creators, can produce visually appealing festival postings in Nepali text.

1.5.1 Scope

Automated Festive Poster Design:

The system will automatically generate aesthetically pleasing poster designs for Nepali festivals (Dashain, Tihar, Chhath, New Year, and Holi) based on text prompts provided in English. It will translate the input text into Nepali script and incorporate it into the poster along with relevant images.

Text-to-Image Synthesis

Producing high-quality images from textual descriptions by applying pre-trained models and adjusting these models so they can comprehend and represent different holiday themes and cultural quirks with accuracy.

Scalability and Efficiency

The system can handle large-scale usage, particularly during peak festive seasons optimizing computational resources to maintain efficiency and responsiveness.

1.5.2 Applications

The Festive Poster Design project has broad applications across various domains.

Personal Use

This system enables users to design their own holiday cards and posters to send via email, publish on social media, or print off in physical cards.

Business and Corporate Use

It enables companies to create personalized holiday posters for staff greetings, marketing initiatives, and promotional materials assisting businesses in maintaining cultural relevance and engagement.

Educational Institutions

Help academic institutions rank and analyze student resumes for co-ops, internships, and post-graduation jobs.

2 LITERATURE REVIEW

Angela Fan et al. [1] depict the Many-to-many MMT model that translates between any pair of languages. It can directly translate from 100 languages to 100 languages or 9900 direction combining dense and sparse language-specific parameters to translate directly between languages. The underlying dataset for M2M-100 was mined from CommonCrawl using a novel strategy that exploits language groupings. It has 7.5B training sentences for 100 languages. The model has the size of 15.4B parameters. Like other state-of-the-art machine translation systems, M2M-100 uses a subword segmentation method and a Transformer-based encoder-decoder architecture. The encoder transforms the source token sequence into a sequence of embeddings of the same length, while the decoder produces the sequence of target tokens. Similarly, the input and output of translation systems are sequences of tokens. These tokens are units from a dictionary built to reconstruct any sentence in any language. Moreover, it integrates a mixture-of-experts approach which allocates different subsets of parameters to specific language groups and enhances the model's ability to handle diverse linguistic features The M2M-100 model outperforms a lot of English-Centric multilingual models by improved BLEU scores. It offers faster and more accurate translations by translating directly between languages without using English as a translator. Its scalable architecture also allows several languages and translation orientations to be supported without increasing the training difficulty.

Jonathan Ho et al. [2] proposed the DDPMs whose methodology involves two primary processes: the forward diffusion process and the reverse denoising process. In the forward process, Gaussian noise is progressively added to the data, transforming it into a nearly Gaussian distribution. This process is mathematically designed to ensure a smooth transition from the original data to noise. The reverse process involves training a model to denoise the data step-by-step, effectively reversing the diffusion process. This denoising training process involves optimizing a simplified variational bound objective, while the sampling process uses a trained model to iteratively remove noise from a Gaussian starting point to generate new data samples. On the CIFAR10 dataset, DDPMs achieve an Inception score of 9.46 and a state-of-the-art Fréchet Inception Distance (FID) score of 3.17, surpassing many existing models including StyleGAN2 + ADA (v1).

Bowen Zhang et al. [3] proposed a transformer-based GAN model named StyleSwin for generating high-resolution images. The generator part of GAN uses a Swin transformer. To generate more accurate images, they proposed double attention which covers contexts of both local and shifted windows. They chose AdaIN for the style injection process. They trained and validated the model on various datasets like CelebA-HQ, LSUN Church, and FFHQ. They evaluated the quality of the image based on the FID score. During the training they used Adam solver with beta1 0.0 and beta2 0,99, They set a 5e-5 learning rate for the generator and 2e-4 for the discriminator. They used standard non-saturating GAN loss with an R1 gradient penalty. They used 8 32GB V100 GPUs to fit 32 images as one training batch for 256 x 256 resolution and only 16 images for 1024 x 1024 resolution. They augmented their data as transformers need large datasets which may lead to overfitting. This model with less computational resources performs better than StyleGAN on the CelebA-HQ 1024 dataset achieving an FID of 4.43 and approaches the performance of StyleGAN2 on FFHQ-1024 achieving an FID of 5.07. This model achieved state-of-the-art quality in generating 256 x 256 images and can also be used for high-resolution images of 1024 X 1024.

Shunan Guo et al. [4] developed a system named Vinci that automatically generated advertising posters using deep generative model Sequence-to Sequence Variational Autoencoder (VAE). The poster was generated based on a product image and many taglines. They also added an online editing feature that allows users to edit the generated poster and provide feedback. This model suggests that it requires relatively less training data compared with GAN based techniques to generate high quality posters. The image features were extracted using VGG-16 pretrained on the ImageNet. This system uses human-designed posters that are stored in a layered format like PSD format to extract design elements. To generate the training samples they labeled 3227 posters generated from Vinci automatically. For training, they collected and generated 173 posters of Chinese seafood and soft drinks. Similar to this system there is the Luban system and they have made comparisons with that system. Further work can be done on this system like diverse datasets and various font face designing and creating more complex systems to generate designs like infographics.

A. Ramesh et al. [5] proposed a two-stage model. Firstly, text captions are encoded into CLIP (Contrastive Language-Image Pre-training) embeddings using a prior model.

Secondly, a decoder generates an image conditioned on image embeddings. A generative model of images given captions, P(x|y) is formed by combining these components. Autoregressive prior and diffusion prior models are explored. Diffusion prior showed better performance and computational efficiency. This approach preserves both semantics and style by making use of robust image representations learned by CLIP. Additionally, 3.5 billion parameter GLIDE model is used as the decoder to produce high-resolution images, and a diffusion upsampler model is used for image upscaling. The models are trained on a combination of CLIP and DALL-E datasets. However, the paper identifies several limitations including attribute binding issues, challenges with details in complex scenes. Moreover, CLIP embedding does not precisely encode spelling information of rendered text.

R. Rombach et al. [6] proposed a model called the Latent Diffusion model. The Latent Diffusion model achieved a new state-of-the-art (SOTA) score for class-conditioned image generation and image inpainting. This also reduces computational complexity, unlike other diffusion models. Generating images using the Latent Diffusion model involves denoising the noise sample multiple times to generate high-resolution images that resemble the training data. The model conditions this process of generating images using input prompts. A text-to-image model with 1.45 billion parameters is trained using language prompts from the LAION-400M dataset. It utilizes a BERT-based tokenizer and a transformer to generate a latent code, which is then integrated into a UNet architecture using cross-attention. The use of cross-attention in the Latent Diffusion model enhances the ability of the model to process the diverse input information effectively for conditiondal image synthesis. The models were trained on an input resolution of 256 x 256 using an A100 GPU but can generalize to larger resolutions like 512 x 512 and 1024 x 1024.

D. Huang et al. [7] proposed a saliency-free poster generation system based on the self-supervised end-to-end neural network by dividing the process into two sub-tasks: layout prediction and attributes identification. The overall objective of this project was to minimize the sum of losses of the two sub-tasks. Over 6000 semi-structured posters were collected for this project out of which 4600 posters were used in 7:1:2 ratio for training, validation, and testing. The dataset contained varying topics like portraits, animals, food, scenes, etc. AdamW optimizer was used to train the entire framework.

The system accepts background image and text as input to generate a poster. Multi-modality feature extraction net (MFEN), an encoder-decoder architecture, was utilized to extract image feature, using a convolution-based deep neural network, and text features, using a bert-base-uncased model, to aggregate them using a decoder to generate a density map of the same size as the original image. The density map was combined with original visual features for attribute identification. The position and size of textual box was predicted by solving an optimization problem on the density map. The model was tested against various saliency-based models, human generated posters and a series of varieties of AuPoD frameworks on test loss, Jaccard similarity, macro-averages accuracy and manual rating where AuPoD proved to be better than saliency-based method but still lacked some stability compared to human generated output. The posters generated by the AuPoD framework had an acceptable rate of 94%, followed symmetry property and implicitly constructed aesthetic constraints.

Ming-Tao et al. [8] present the challenges in generating high-quality images from textual descriptions. Traditional GANs used for text-to-image synthesis led to several issues that introduced entanglements between generators of different image scales, limited supervision due to fixed extra networks, and computationally expensive cross-modal attention restricting its use to specific scales. Meanwhile, the DF-GAN (Deep Fusion Generative Adversarial Networks) model provides advanced text-to-image synthesis, offering a straightforward effective approach. DF-GAN overcomes the limitations of traditional GANs by introducing three key innovations. It includes a One-Stage Backbone, Target Aware Discriminator, and Deep-Text Image Fusion Block. One-Stage backbone synthesizes high-resolution images directly avoiding entanglement issues of multi-stage architecture. Similarly, Target-Aware Discriminator improves semantic consistency without the need of extra networks by using a Matching-Aware Gradient Penalty (MA-GP) and one-way output ensuring smooth loss surface at actual and matched data points. DFBlock enhances the fusion of textual and visual elements, allowing for a more thorough integration of text information at different image sizes. However, DF-GAN produces sentence-level text information restricting detailed visual feature synthesis. This issue can be addressed by introducing pre-trained LLMs.

A. Sauer et al. [9] combined the StyleGAN architectures with CLIP embeddings, for text-to-image synthesis. StyleGAN-T attempts to improve semantic alignment between text descriptions and generated images by embedding text prompts using a pre-trained CLIP text encoder and feeding these embeddings into both the generator and discriminator. This paper highlights the computational drawbacks of translational equivariance which is not necessary in text-to-image synthesis, choosing instead to use the more effective StyleGAN2 backbone to lower computing expenses and make it easier to introduce stochastic fluctuations in low-level features by using output skip connections and spatial noise inputs. It introduces two changes to the generator architecture: an easily optimizable residual block to normalize input and scale the output, and stronger conditioning that improved the FID and CLIP score by ~10%. The discriminator has been redesigned with a ViT-S for the feature network. Furthermore, guiding the text encoder and the generator and explicit truncation further boosted the quality and alignment for a given sample. However, while CLIP-based guidance enhances text alignment, it may cause artifacts in images; hence, it may be necessary to retrain CLIP on data with better resolution to address these problems. In addition, the paper notes persistent issues such attribute binding and text coherence within images and suggests future options for further conditioning mechanism enhancements.

J. Lin et al. [10] proposed an automated poster generation system that takes product images and product descriptions and input to produce posters of varying sizes as the result. The system consisted of four key stages: image cleaning and retargeting, layout generation, tagline generation, and style attribute prediction. Image cleaning involves erasing graphic elements on the product image while image retargeting involves adjusting image size. The methods utilized for the first step were graphic element detection, inpainting, outpainting, saliency detection, cropping, and scaling. The second step, layout generation, utilized transformer-based CGL-GAN (Composition-aware Graphic Layout GAN) and ICVT to generate a content-aware poster layout. The tagline generation step used Content-driven Dense-Captioning on Images, a multi-modal generative model. Style attribute predictor (SAP), a transformer-based model, could simultaneously predict font typography, dominant color, stroke color, and gradient color of taglines. 62 commonly used font typography was utilized for tagline font. The project also contributed a dataset of 76,960 posters annotated for tagline content, graphic element position, style attributes, and font typography. A font-weight-aware font

classifier using a self-supervised approach was trained for font typography. Upon comparison with MMCQ, Retrieval, and Classifier methods, AutoPoster produced better R1, R1+R2 scores along with better D-ab and D-light metrics. Additionally, upon comparison with Luban and posters designed by professionals on a voting basis consisting of 129 participants, 20 professionals, and 109 non-professionals, AutoPoster exhibited higher R1 (78%) and R1+R2 (58%) metrics. Furthermore, AutoPoster was compared with posters generated by professionals on 80,000 products where AutoPoster had a higher CTR (click-through rate) and RPM (revenue per mile) increase of 8.24% and 8.25%.

Table 2-1 Summary Table of Literature Review

Title	**Methodology**	**Inferences**
Beyond English-Centric Multilingual Machine Translation (21 Oct, 2020) **Author: Angela Fan et al.**	The model uses transformer architecture with dense scaling and language specific sparse parameters.	• Allows direct translation between any pair of 100 languages. • Achieves over 10 BLEU points higher than English-centric models for non-English language pairs. • Reduces biases from using English as an intermediary and handles large-scale translations efficiently.
Denoising Diffusion Probabilistic Models (DDPM) (16 Dec, 2020) **Author: Jonathan Ho et al.**	Uses a forward diffusion process adding Gaussian noise to data and a reverse denoising process to generate data by removing the noise.	• Achieves high-quality image generation with stability and diversity, leveraging a probabilistic framework for better mode coverage than GANs. • Computationally intensive; requires large amounts of data, GPU resources and time for training.

StyleSwin: Transformer-based GAN for High-resolution Image Generation (21 Jul, 2020) **Author: Bowen Zhang et al.**	The generator model is a Swin Transformer and the Discriminator model is a wavelet discriminator.	• Resolution: 256 x 256 SOTA • To generate more accurate images, they proposed double attention. • Less computational resources than StyleGAN.
Vinci: An Intelligent Graphic Design System for Generating Advertising Posters (2021) **Author: Shunan Guo et al**	The model uses a deep generative model Sequence-to Sequence Variational Autoencoder (VAE).	• Takes an input image and extracts image features using VGG-16 pre-trained on the ImageNet. • Has an online editing feature for feedback. • Requires less training data and computational resources than GANs.
Hierarchical Text-Conditional Image Generation with CLIP Latents (13 April, 2022) **Author: Aditya Ramesh et al.**	Uses a two-stage model combining CLIP embeddings with diffusion models. Also, Experiments with both autoregressive and diffusion models for the prior, found diffusion models more efficient and better in terms of sample quality.	• Achieves high-quality image generation in zero-shot settings on the MS-COCO dataset. • Human evaluators significantly prefer this model's outputs over prior state-of-the-art methods.Computationally intensive and requires significant resources for training.
High-Resolution Image Synthesis with Latent Diffusion Models (13 April, 2022) **Author: Robin Rombach et al.**	The model denoises the noise samples multiple times to generate high-resolution images. Also, the model uses cross-attention layers to allow conditioning input.	• Achieves State of the art for class-conditional image synthesis and image inpainting. • Latent-based diffusion model • Less computational requirements than pixel-based diffusion models.
AUPOD: End-to-End Automatic Poster Design by	The model is a self-supervised layout prediction and attributes identification	• Density mapping for layout prediction

Self Supervision (28 Apr 2022) **Author: Dongjin Huang et. al.**	model trained in semi-structured posters. The image encoder is a convolution-based deep neural network and the text feature encoder is a pre-trained contextual token embedding. It uses multilevel perceptron (MLP) to convert textual representation to vector space	• Uses a unified neural network for layout prediction and attributes identification. • Performs better than the saliency-based method on layout prediction sub-task. • Higher variance is score than manually generated posters
DF-GAN: A Simple and Effective Baseline for Text-to-Image Synthesis(15 Oct,2022) **Author: Ming Tao et al.**	Target-Aware Discriminator with Matching-Aware Gradient Penalty and One-Way Output	• Outperforms state-of-the-art models (e.g., AttnGAN, DM-GAN) in image quality and text-image semantic consistency. • Achieves better IS and FID on CUB and COCO datasets. • Simpler and more efficient with fewer parameters and no extra networks.
StyleGAN-T: Unlocking the Power of GANs for Fast Large-Scale Text-to-Image Synthesis (23rd Jan, 2023) **Author: Axel Sauer et. al**	Generator with residual convolutions and stronger text conditioning, while the discriminator employs a ViT-S feature network and applies augmentation techniques to improve performance. It also employs CLIP-based guidance during training to align generated images closely with textual descriptions.	• Outperforms distilled diffusion models at lower resolution in terms of quality and speed. • Computationally intensive, requiring significant resources for training and inference.
AutoPoster: A Highly Automatic and Content-aware Design System for Advertising Poster Generation (23 Aug 2023) **Author: Jinpeng Lin et. al.**	It utilizes CGL-GAN and ICTV for layout generation. Tagline generation utilizes multi-modal generative model. Style Attribute Predictor (SAP) has CNN based encoder and self-attention layer and cross-attention based decoder.	• Higher R1 and R1+R2 metrics than Luban and posters designed by designers.

3 METHODOLOGY

This section provides the detail working of the system and the different models used.

3.1 System Architecture

This diagram shows the overall working of the system.

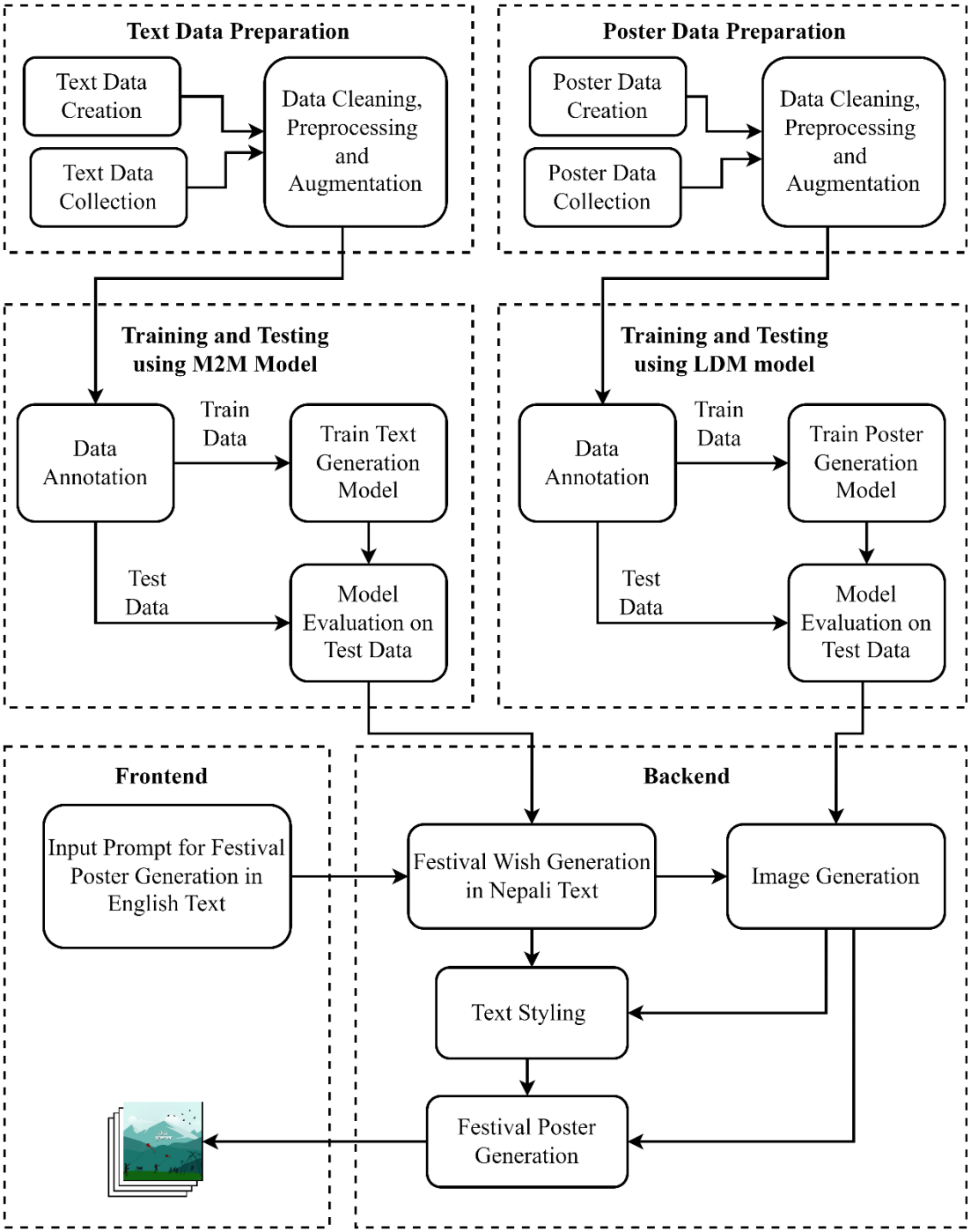


Figure 3-1 System Architecture

3.1.1 Working Principle

The system asks the user to input a prompt in the English language. A fine-tuned M2M Model is used to generate festival wishes in the Nepali language. The model also extracts the festival theme from the given input prompt. Simultaneously, a fine-tuned Latent Diffusion Model is used to create a relevant image suitable for the poster based on the identified theme.

Once the image and text are generated, the ColorThief library is applied to identify dominant colors from the generated image which is then used to determine the ideal text color. At the same time, saliency mapping is employed to determine the ideal placement of styled text within the generated image. Font style is randomly generated from existing fonts available in the database. Finally, PIL, a Python Imaging Library, is utilized to seamlessly integrate the generated image with the styled text image to generate a complete festival poster which is then presented to the user.

3.2 Dataset Preparation

3.2.1 Dataset for Title Generation in Nepali Font

The dataset for title generation will be collected manually from different sources, such as social media posters, greeting cards, festival messages, and other relevant Nepali literature. The dataset will contain text prompt in English and its corresponding wishes or greetings text in Nepali font. If needed the data is augmented to increase the dataset size.

For example:

Prompt: "I want a poster to wish happy Dashain to Nepali people for the year 2079."

Text: "२०७९ सालको दशैंको हार्दिक मंगलमय शुभकामना!"

This pair will be stored in a structured format such as a CSV file for easy access under two labels ‘Prompt’ and ‘Text’. The text prompt as input is cleaned by removing stop words and tokenized to create a cleaned dataset. Finally, the data is ready to be fine-tuned on a pre-trained M2M Model.

3.2.2 Dataset for Poster-Generation in Nepali Font

The project aims to create a dataset for training a title generation model focused on five Nepali festivals: Dashain, Tihar, Chhath, New Year, and Holi. The dataset will be manually collected from various websites like Pinterest, Facebook, and Dribble, featuring festival-related images with styled Nepali text. Before inclusion, all non-essential elements like text, logos, and other information will be removed using inpainting tools. The image resolution will then be enhanced to ensure high quality.

To augment the dataset and improve model training, various techniques will be applied. Each image in the dataset will be annotated with tags indicating the festival theme and the corresponding Nepali title text. This annotated dataset will be essential for training generative models effectively, enabling them to generate accurate and contextually appropriate titles for festival-related images.

(a) Raw Image for Tihar Dataset

(b) Image for Tihar dataset after removing text

(c) Image for dataset after removing text and enhancing resolution

Figure 3-2 Dataset Preparation for Tihar Festival

3.3 Title Generation

Title generation is the first and crucial phase of our system. This stage ensures the final poster conveys the intended message and sentiment appropriately. Firstly, the user enters the prompt in English for generating festival posters. The input prompt describes the desired message or wish for a specific festival. Once the prompt is entered, our system processes this input to generate a coherent, logical, and well-contextualized Nepali title or wish that will be kept in the poster. This involves understanding the context of the input prompt by identifying the festival and extracting the key message or wish.

The system will use the M2M-100 model to generate wishes in Nepali language based on the input prompt.

The M2M-100 model is a state-of-the-art multilingual machine translation developed by Facebook AI in 2020. The M2M model offers direct translation between 100 languages. It's based on the Transformer model and uses shared embeddings, which helps it understand different languages better. With extensive training and advanced techniques like back-translation, it can generate accurate titles directly from one language to another without translating through English or any middle language, preserving context and meaning. Therefore, we've decided to use the M2M-100 model given its promising capabilities of performing various translation tasks effectively.

3.3.1 M2M-100 Architecture

M2M-100 is a multilingual sequence-to-sequence model trained for Many-to-Many multilingual translation. It is composed of two modules: 12-layer encoder and 12-layer decoder. The encoder processes the source language input, and the decoder generates the target language output. It contains 8192 Feed-Forward-Network size and 1024 embedding dimensions. The total parameter count is 1.2B.

The model relies heavily on Cross-attention and self-attention mechanisms. Cross-attention links the encoder and decoder, allowing the model to concentrate on pertinent segments of the input sequence when producing the output. Self-attention enables the model to assess the relative relevance of various words in a sentence. Additionally, it uses shared vocabulary that allows it to learn cross-lingual representations. This shared vocabulary is typically constructed using techniques like Byte Pair Encoding (BPE) or SentencePiece to handle the diverse character sets and token frequencies of different languages.

The encoder takes the sequence of tokens $W = (w_1, \ \ldots w_s)$ and the source language l_s and produces a sequence of embeddings of the same length $H = (h_1 \ \ldots h_s)$:

$$H \ = encoder(W, l_s) \qquad 3.1$$

Then, the decoder is fed the sequence of embeddings H with the target language l_t and sequentially produces the target sentence token by token $V = (v_1, \ldots \ v_s)$:

$$v_{i+1} = decoder(H, l_t, v_1 \dots v_i) \quad 3.2$$

Encoder and Decoder are composed of the same layers called Transformer layers. Transformer layers are composed of two sublayers, self-attention layer and feed-forward layer. The self-attention layer is an attention layer that updates each element of the sequence by looking at the other elements and the feed-forward layer (FFN) passes each element of the sequence independently through a 2-layer MLP.

$$Z = norm\big(X + self - attention(X)\big) \quad 3.3$$

Where, X is input to the first self-attention layer and $self - attention(X)$ is self-attention applied to X.

$$Y = norm\big(Z + self - attention(Z)\big) \quad 3.4$$

Where, Z is output after the first self-attention and $self - attention(Z)$ is self-attention operation applied to Z.

The Transformer architecture has been designed for the bilingual case, where the target language is fixed. In the case of multilingual machine translation, the target language is not fixed, so they added a special token in the encoder indicating the source language l_s and a special token in the decoder indicating the target language l_t.

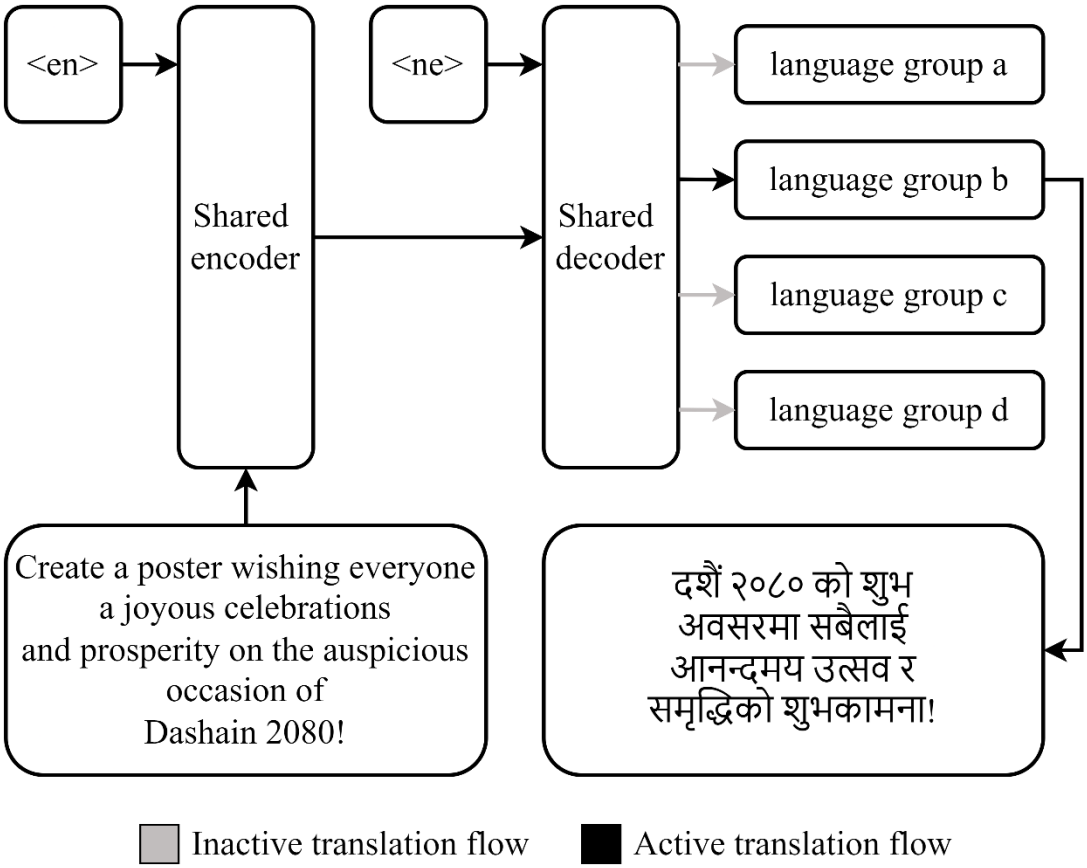


Figure 3-3 English to Nepali Text with Dense and Language Specific Sparse Model

The model is trained on a vast multilingual dataset that includes parallel sentences from many language pairs with the Adam optimizer and warmed-up first for 4000 updates, with label smoothing 0.1. For regularization, the dropout parameter is tuned between {0.1, 0.2, 0.3}. To stabilize the training of deeper Transformers, it is trained with LayerDrop 0.05 and pre-normalization.

Through extensive training on diverse data and using techniques like dropout and regularization, the model is made robust to various linguistic phenomena and can generalize well to new inputs.

3.3.2 Evaluation Metrics

BLEU Score

BLEU, or the Bilingual Evaluation Understudy, is a score for comparing a candidate translation of text to one or more reference translations. It measures the overlap between

n-grams in the generated text and reference text, indicating how similar the generated text is to the reference text.

The formula for BLEU score is as follows:

$$BLUE = BP \times exp(\sum pn) \qquad 3.5$$

Where, BP(Brevity Penalty) is a penalty term that adjusts the score for translations that are shorter than the reference translations and pn is the precision of n-grams.

BLEU score ranges from 0 to 1, with higher values indicating better translation quality.

ROUGE Score

ROUGE score measures the similarity between the machine-generated summary and the reference summaries using overlapping n-grams, word sequences that appear in both the machine-generated summary and the reference summaries, focusing on recall and precision of n-grams.

The formula for ROUGE score is as follows:

$$ROUGE = \sum(Recall\ of\ n - grams) \qquad 3.6$$

Where, $Recall\ of\ n - grams$ is the number of n-grams that appear in both the machine-generated summary and the reference summaries divided by the total number of n-grams in the reference summaries.

ROUGE score ranges from 0 to 1, with higher values indicating better summary quality.

3.4 Image Generation

Once the theme is extracted from the input prompt, a new image specific to the theme is generated using a pre-trained Latent Diffusion Model. The Latent Diffusion model is a popular open-source diffusion model that achieved a new state-of-the-art (SOTA) score for class-conditioned image generation. This also reduces computational complexity, unlike other diffusion models. Generating images using the Latent Diffusion model involves denoising the noise sample multiple times to generate high-

resolution images that resemble the training data. The model conditions this process of generating images using input prompts. It involves two training steps: Training the autoencoder and then training the diffusion model. Autoencoder consists of an encoder and a decoder. The autoencoder is trained once and then there is no need to train the encoder and decoder again in the diffusion model. Then the diffusion model is trained to denoise the noising samples. The use of cross-attention in the Latent Diffusion model enhances the ability of the model to process the diverse input information effectively for conditional image synthesis.

3.4.1 Latent Diffusion Model

The latent diffusion model addresses the computational challenges of training diffusion models to generate high-resolution images by separating the compressive and generative learning phases. This separation is achieved by introducing an autoencoding model that learns a lower-dimensional space representing the input data in a perceptually equivalent manner to the original image space.

- The computational complexity is significantly reduced by operating in a low-dimensional space learned by an autoencoder.
- Utilizing the inductive bias from the UNet architecture in diffusion models enhances their effectiveness, especially for data with spatial structure.
- The proposed method yields general-purpose compression models with a latent space for multiple purposes. This latent space can be used to train diverse generative models and for various downstream applications, including tasks like single-image CLIP-guided synthesis.

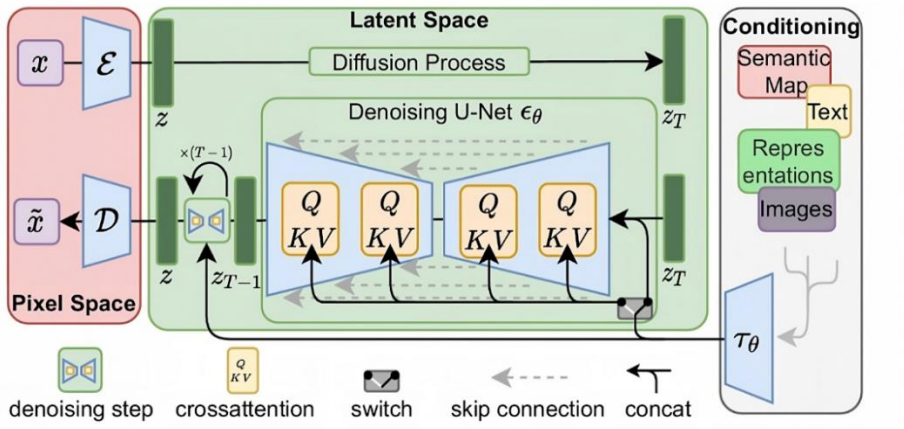


Figure 3-4 Architecture of Latent Diffusion Model [6]

Perceptual Image Compression

The perceptual image compression has an autoencoder. Combining perceptual loss and patch-based adversarial objective trains the autoencoder. This strategy keeps the reconstructed images natural and realistic, avoiding blurriness that occurs with pixel space losses like L2 (Mean Square Error) or L1 (Mean Absolute Error).

Perceptual Loss

Perceptual loss is a loss function that measures the difference in the high-level features. These features are extracted from a pre-trained neural network (usually a convolutional neural network like VGG).

Patch-based Adversarial Objective

The patch-based adversarial objective is used in GANs to ensure local realism in generated images. It divides the image into smaller patches, and a discriminator evaluates each patch independently.

Given an input image $x \in \mathbb{R}^{H \times W \times 3}$ in RGB space:

Here, H represents height, W represents weight and 3 is the RGB value for each pixel.

Encoder (E) encodes x into a latent representation z.

$$z = E(x) \tag{3.7}$$

Decoder (D) reconstructs the image from the latent representation.

$$\bar{x} = D(z) = D\big(E(x)\big) \qquad 3.8$$

Where $z \in \mathbb{R}^{H \times W \times 3}$

The encoder downsamples the image by dividing it by $f = \frac{H}{h} = \frac{W}{w}$. Different downsampling factors $f=2^m$ with $m \in \mathbb{N}$ are explored.

To avoid high-variance latent spaces, two types of regularizations are experimented with:

- KL Regularization (KL-reg.): Applies a slight KL-divergence penalty towards a standard normal distribution on the learned latent space, similar to a Variational Autoencoder (VAE).
- Vector Quantization Regularization (VQ-reg.): Incorporates a vector quantization layer within the decoder, analogous to VQGAN but with the quantization layer integrated into the decoder.

Since this Diffusion model works with two-dimensional learned latent space z=E(x), allowing for relatively mild compression rates and very good reconstructions. As a result, our compression model preserves the details of the original image better.

Diffusion Models are probabilistic models that learn data distribution by denoising a normally distributed variable, effectively reversing a fixed Markov Chain process. They use a reweighted variant of the vibrational lower bound p(x), similar to denoising score-matching, and consist of a sequence of denoising autoencoders $\epsilon_\theta(x_t,t)$ predicting cleaner versions of their inputs. The objective simplifies to a denoising loss term, allowing efficient training in a low-dimensional latent space that captures essential details.

$$L_{DM} = E_{x,\epsilon\sim N(0,1),t}[\|\epsilon - \epsilon_\theta(x,t)\|]^2 \qquad 3.9$$

With t uniformly sampled from {1,.....,T}.

With our trained perceptual compression models, we have an efficient, low-dimensional latent space where high-frequency, imperceptible details are abstracted away. This space is more suitable for likelihood-based generative models because they can (i) focus on important semantic information and (ii) train more efficiently in a lower-dimensional space.

Unlike previous models that used autoregressive or attention-based transformers in a compressed latent space, our approach leverages image-specific inductive biases. This includes using 2D convolutional layers in the UNet and focusing on perceptually relevant details with a reweighted objective.

$$L_{DM} := E_{E(x),\epsilon\sim N(0,1),t}[\|\epsilon - \epsilon_\theta(z_t, t)\|]_2^2 \qquad 3.10$$

The model's neural backbone, ϵ_θ (o, t), is a time-conditional UNet. During training, the latent variable z_t is efficiently obtained from encoder E. Samples from the prior distribution p(x) can be decoded back to image space with a single pass through decoder D, simplifying and speeding up the generation process.

Conditioning Mechanisms

To make the diffusion models more flexible in generating conditional images, a cross-attention mechanism is integrated into the UNet backbone. This mechanism enhances the model's ability to handle text prompts. A domain-specific encoder $\mathcal{T}_\theta$ preprocesses these inputs into an intermediate representation $\mathcal{T}$ (y) in $\mathbb{R}^{Mxd_r}$ dimensions. This representation is then utilized by a cross-attention layer within the UNet, implementing the Attention mechanism. This mechanism computes softmax $\left(QK^T/\sqrt{d}\right)$. V, where Q, K, and V linear transformations of the latent variable z_t, the encoded input $\mathcal{T}_\theta(y)$, and the intermediate representation, respectively. Learnable projection matrices facilitate this mapping process, enabling effective integration and processing of diverse input modalities for conditional image synthesis.

Based on pairs of images and conditioning inputs, we train the conditional Latent Diffusion Model (LDM) using the following objective:

$$L_{DM} := E_{E(x,y),\epsilon\sim N(0,1),t}\left[\left\|\epsilon - \epsilon_\theta\left(z_t, t, \mathcal{T}_\theta(y)\right)\right\|\right]_2^2 \qquad 3.11$$

Where, ϵ is sampled from a standard normal distribution, is the latent variable obtained from the encoder E, ϵ_θ is the output of the conditional denoising autoencoder dependent on z_t, diffusion step t, and the intermediate representation $\mathcal{T}_\theta(y)$ of the conditioning input y.

Both $\mathcal{T}_\theta$ and ϵ_θ are optimized together according to this objective. The conditioning mechanism is adaptable, allowing $\mathcal{T}_\theta$ to be parameterized with domain-specific models like transformers, particularly useful when conditioning inputs y are text prompts.

3.4.2 Evaluation Metrics

Frechet Inception Distance (FID)

The Frechet Inception Distance score is a metric that calculates the distance between feature vectors calculated for real and generated images. Lower FID scores indicate that the generated images are closer to the real images in terms of distribution, which typically corresponds to higher quality and more realistic generated images.

Let (M_t, C_t) and (M_g, C_g) represent the mean and covariance of the true and generated features respectively, then we compute

$$FID = \|M_t - M_g\|_2^2 + Tr\left(C_t + C_g - 2\left(C_t C_g\right)^{\frac{1}{2}}\right) \qquad 3.12$$

Inception Score (IS)

It evaluates the quality and diversity of generated images based on classification confidence from a pre-trained Inception network. The IS takes a list of images and returns a single floating point number, the score which is the measure of how realistic the output is. A higher score is better.

Let, g(z) be an image generator to be evaluated, y be a label, p(y|x) be the posterior probability of a label computed using the Inception V3 model on image x.

Let the marginal class distribution be,

$$p(y) = \int p\big(y|g(z)\big)dz \qquad 3.13$$

Similarly, D(p||q) be the KL-divergence between two probability distributions p and q. The Inception Score for a generator is:

$$IS = exp\left[E_{z\sim p}(z)\left[D\left(p\big(y|g(z)\big)||p(y)\right)\right]\right] \qquad 3.14$$

Qualitative Metrics

Qualitative metrics are descriptive indicators based on subjective assessment.

Human Evaluation: Subjective assessment by human evaluators on criteria such as fluency, coherence, and relevance of the generated text.

Aesthetics Score: Volunteers or experts rate the quality, relevance, and aesthetics of the generated posters and how well the generated layouts match the provided textual descriptions. Volunteers score the generated posters on a certain scale on the aesthetics of the layouts then the mean and the standard deviation of the scores are taken.

3.5 Text Styling

The festival wish in the Nepali language generated by the M2M model is transformed into an aesthetically pleasing styled text using a text styling algorithm. The algorithm determines the font type, font colour, font size, and position of text. ColorThief, a Python package is used to extract the dominant color i.e. background color and six best palettes from the image generated by the LDM. The colours are then used to determine the text colour by ensuring there is enough contrast between the text colour, which is randomly extracted within the image colour palette, and the background colour. The text font type is selected randomly from a collection of fonts stored in the database. Saliency mapping technique is used to determine font size and text position on the festival post. Once all the features are determined, a styled image picture is generated which is to be placed on the background image.

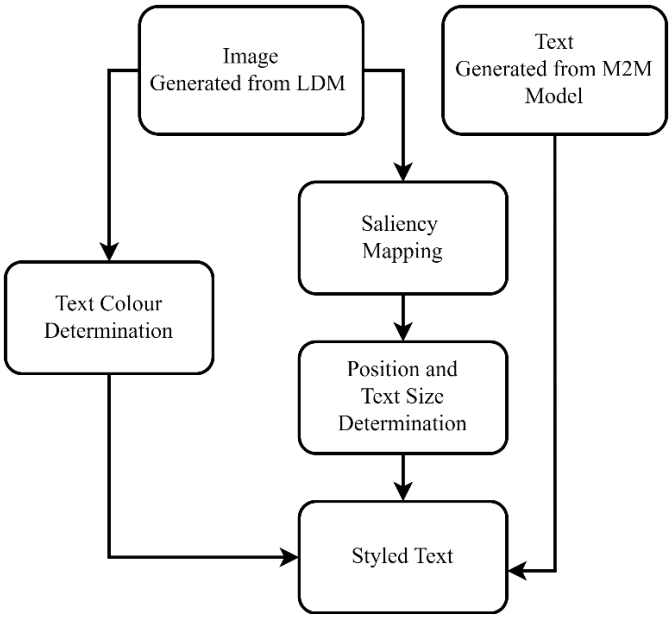


Figure 3-5 Text Styling Procedure for Generated Text

3.6 Text-Image Integration

The background image, generated by the LDM, and the styled text image, generated using a text styling algorithm, are integrated into a single festival post using the PIL library. The styled text is placed at the position determined by saliency mapping technique.

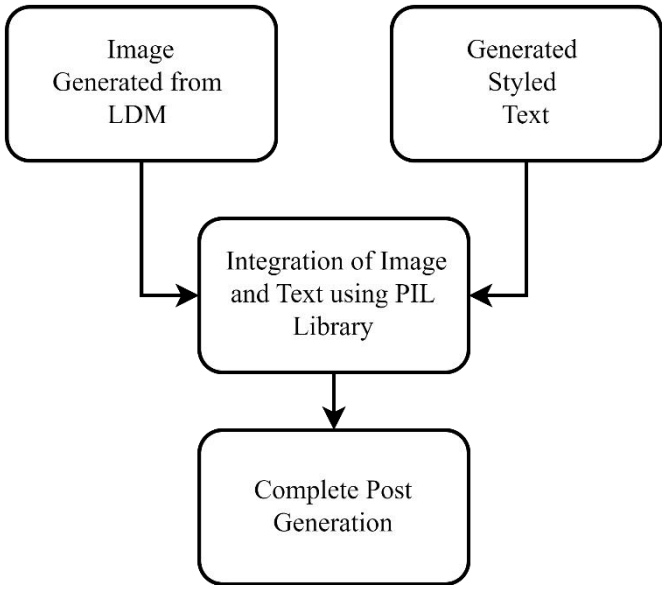


Figure 3-6 Integration of Generated Image and Styled Text

3.7 Flowchart

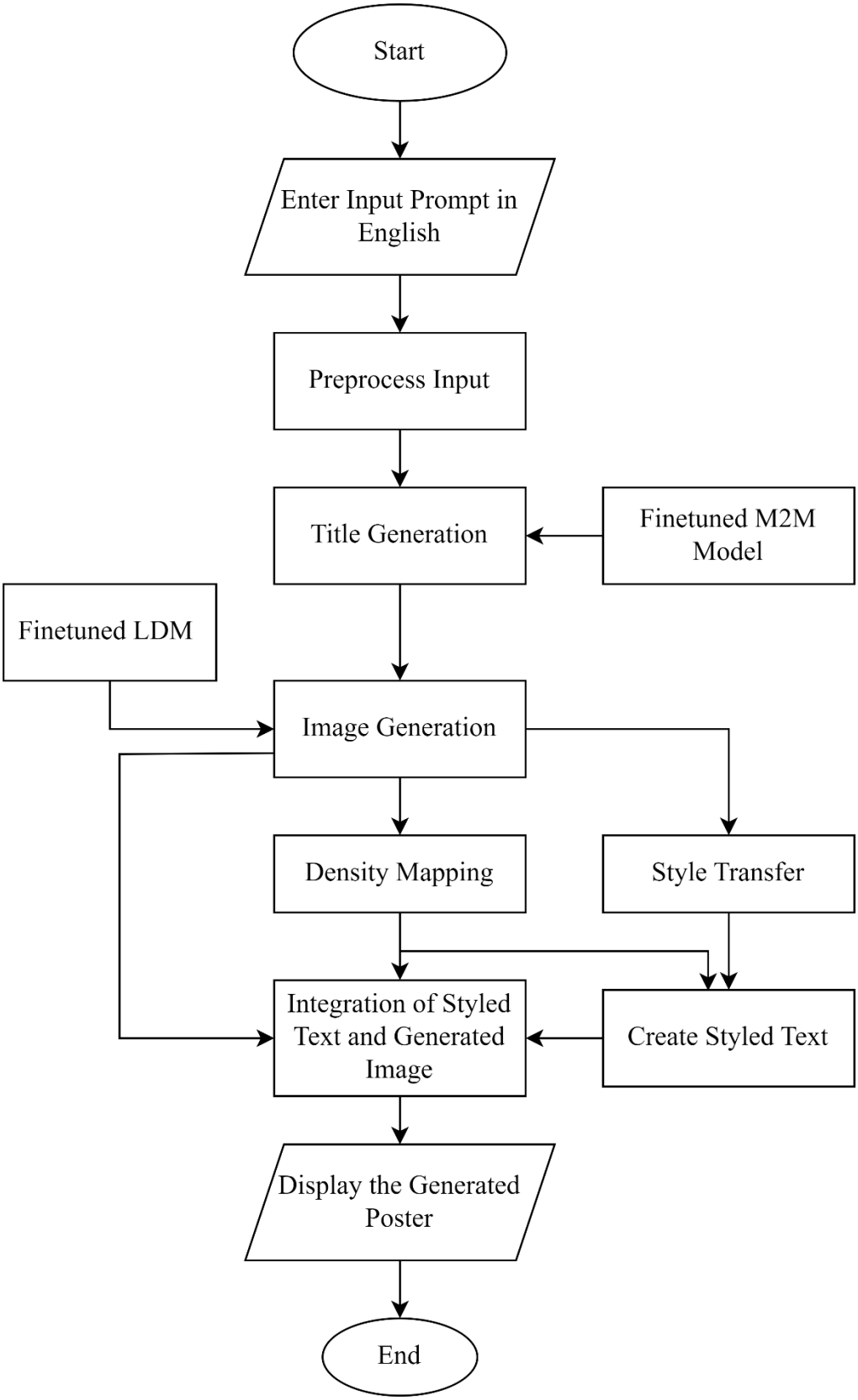


Figure 3-7 System Flowchart

3.8 Instrumentation Tools

3.8.1 System Requirements

Operating System, GPU Server and Memory

We need a computer or laptop with any OS Windows, Linux, or Mac to develop our system. Google Colab provides free access to GPUs, including the NVIDIA Tesla T4 with up to 16 GB of memory, which is suitable for deep learning and machine learning tasks. Kaggle also offers free access to GPUs, specifically 29 GB of 2 Tesla T4 GPUs and 1 P100 GPU, with a usage limit of 30 hours per week or 9 hours per session.

3.8.2 Software Tools and Library Requirements

VS Code and Google Colab

For coding tasks, our team will use Visual Studio Code (VS Code) as the primary Integrated Development Environment (IDE). VS Code enhances the exploratory and prototyping phases of our project by facilitating a thorough, team-based approach to coding. This allows us to create effective prototypes and iterate efficiently. Google Colab and Kaggle are cloud-based platforms that allow the execution of Python code in a web-based notebook format.

NLTK

Tokenizer is a key component of Natural Language Processing used to segment text into tokens, which can be words, characters, or sentences. We will utilize the NLTK tokenizer for this purpose. This system tokenizes each word from an English input prompt to extract key themes accurately.

HTML, CSS, and Javascript

HTML, CSS, and JavaScript are used together to create a simple, user-friendly interface through which the user can interact. The primary structure of a web page is created with HTML, a standard markup language. HTML components are styled with CSS. It explains how web pages should be laid out. Interactive web pages are created using ascripting language known as JavaScript.

FastAPI

FastAPI is a modern web framework for building APIs with Python 3.7+ that emphasizes performance and ease of use. It utilizes Python type hints to provide automatic interactive API documentation, making it user-friendly and reducing bugs. FastAPI's combination of speed, simplicity, and powerful features makes it an excellent choice for deploying web applications and APIs.

ColorThief

ColorThief is a Python library that is used to extract the single most dominant colour as well as a palette of the most common colours in the image by processing its pixels using algorithms such as the Median Cut algorithm. It is capable of processing images in various formats like jpg, png, etc.

PIL/Pillow

PIL (Python Imaging Library) is a widely used library for opening, manipulating, and saving various image file formats in Python. It adds processing capabilities like point operations, resizing, rotation, and arbitrary affine transforms, allows drawing graphical elements, and offers various filters to Python interpreters. PIL is commonly used for image archival and batch-processing applications. Pillow is a fork of PIL that provides backward compatibility as well as extends the functionality of the original library.

OpenCV

Open Source Computer Vision Library (OpenCV) is used to perform many image-related tasks. For reading image files, we use cv2.imread (image_path). For converting the BGR (Blue-Green-Red) color space of an image to the RGB (Red-Green-Blue) color space, we use cv2.cvtColor(image, cv2.COLOR_BGR2RGB). To display the image, we use the function cv2_imshow(image).

4 EXPECTED OUTCOME

The goal of this project is to generate high-quality, eye-catching posters that accurately reflect the textual descriptions provided. These expectations are detailed below, along with example photographs to demonstrate the expected results.

4.1 Visual Quality

The model will generate posters that have the following characteristics:

Resolution

The generated posters will have a resolution of up to 256x256 pixels, which ensures clarity and detail in the images.

Color Fidelity

Posters will have vibrant colors, enhancing their visual appeal and making them suitable for various applications, including promotional and decorative purposes.

Aesthetic Appeal

The images will be visually appealing and effectively deliver the intended message.

4.2 Relevance and Accuracy

Each poster will be designed according to the respective festival theme that reflects the cultural and contextual significance of the events.

Text-to-Image Synthesis

The model will generate images that precisely reflect the descriptions provided.

For instance, the created poster will include traditional images like people putting tika, flying kites, or playing on swings if the text prompt mentions or describes the Dashain celebration. Similarly, for the Tihar festival, the model will produce posters showcasing decorative lights, rangoli designs, and the worship of animals like dogs and cows.

Contextual Relevance

In order to ensure that the created images are not only aesthetically pleasing but also culturally and contextually relevant, the posters will be in line with the specific context and intended message of the textual descriptions. This implies that the developed poster for each festival will include customs and events related to that particular occasion, making the posters audience-relevant and poignant.

For example:

Text Prompt: "Make a poster for Dashain."

Generated Title: "२०७८ सालको बडा दशैंको उपलक्ष्यमा हार्दिक मङ्गलमय शुभकामना व्यक्त गर्दछौं।"

Generated Poster: The poster may include images of kids playing on traditional bamboo swings (ping), family members placing tika on each other's foreheads, kites soaring in a clear autumn sky, or other similar situations that perfectly encapsulate the essence of Dashain.

Figure 4-1 Poster for Dashain Festival

Prompt: “Create a Dashain poster.”

Prompt: “Make a poster for New year 2081.”

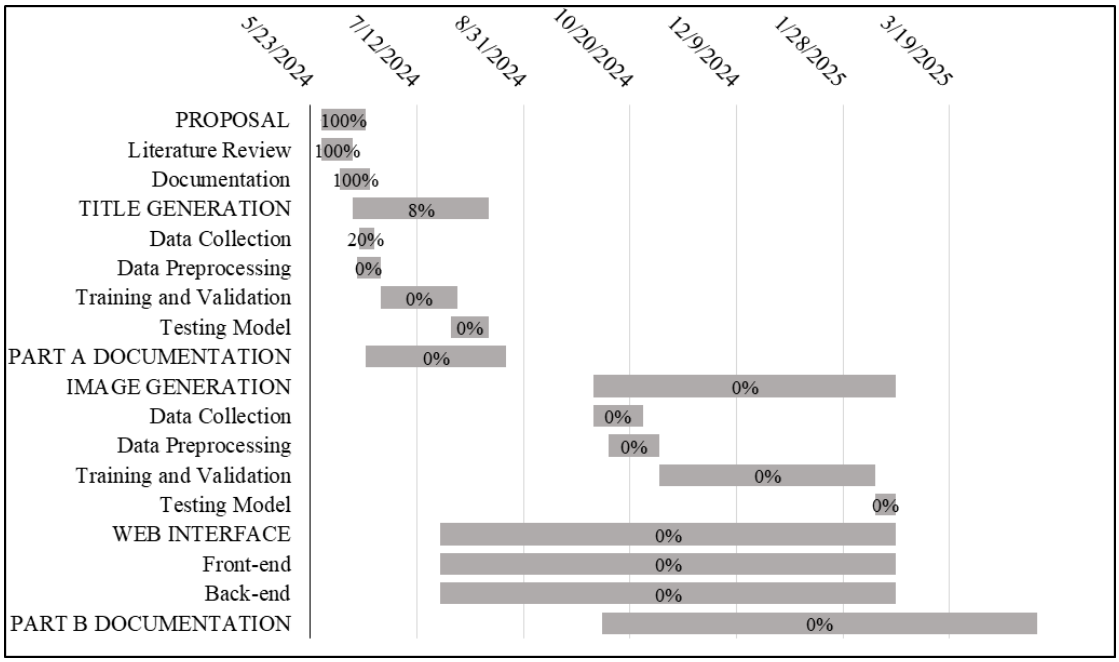
Prompt: “I want a Dashain poster with an appropriate message.”

Prompt: “Give me a poster for chhath puja”

Figure 4-2 Images generated for Respective prompts.

5 PROJECT SCHEDULE

Table 5-1: Project Timeline Gantt chart



6 PROJECT BUDGET

Table 6-1 Budget Estimation

Item	**Price (NPR)**
Printing	5000.00
Colab Resource (1400 per month)	4200.00
Domain (per year)	500.00
Hosting (per year)	2000.00
Grand Total	**11,700.00**

7 FEASIBILITY ANALYSIS

7.1 Technical Feasibility

Our project does not require a high degree of computer specification. Python programming will be used to create the system. The system is operational with the least number of technical requirements and hence, it is technically feasible.

7.2 Operational Feasibility

The Graphical User Interface (GUI) of the system will be user-friendly and interactive, making it easier to provide input prompt.

7.3 Economic Feasibility

Since our project is purely software-based, it is extremely feasible in terms of economy. The project will be developed on a personal computer. However, we have to buy the domain name for deploying the project. If required, we will purchase the GPU resource for training.

7.4 Schedule Feasibility

The project will be completed within the designated timeframe. Our project timeline is structured around key milestones, and we are prepared to address any challenges that may arise to ensure on-time completion.

7.5 Legal Feasibility

The dataset collection process ensures no harm is caused to any festivals or through the use of offensive language or text. Company names or logos in images will be removed during preprocessing. All datasets are sourced exclusively from public websites.

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